

Abstract—The rapid growth of social media has led to an alarming rise in fake profiles used for misinformation phishing identity theft and social engineering attacks. Existing rule-based detection mechanisms are insufficient against increasingly sophisticated bot and fake account behavior. This paper proposes a multimodal AI-based fake profile detection system that combines an ensemble machine learning approach with a Convolutional Neural Network (CNN)-based face detection and analysis module. The ensemble model fuses behavioral linguistic structural and visual (face-based) features to classify social media profiles as genuine or fake. Profile images are analyzed using a CNN-based face detector to extract authenticity indicators such as GAN-generated face probability face count anomalies and facial consistency scores. Experimental evaluation on publicly available datasets demonstrates that the proposed multimodal system achieves an accuracy of 98.1% outperforming text-only and metadata-only approaches. The system further incorporates explainability through feature importance analysis offering transparency in detection decisions.

Keywords—fake profile detection face detection GAN face social media ensemble learning CNN XGBoost Random Forest deep learning

I. Introduction

Social media platforms such as Twitter Instagram and Facebook host billions of users worldwide. However a significant fraction of these accounts are fake profiles bots or sockpuppets created for malicious purposes including spreading disinformation manipulating public opinion conducting phishing attacks and harvesting personal data [1]. According to recent reports Twitter alone suspended over 70 million suspicious accounts in a single month highlighting the scale of the problem [2].

A distinctive and often overlooked indicator of fake profiles is the profile picture itself. Many fake accounts use AI-generated (GAN-synthesized) faces stock photographs or stolen images as profile pictures to appear authentic. Generative Adversarial Networks (GANs) such as StyleGAN2 can produce photorealistic human faces indistinguishable to the naked eye making visual inspection alone insufficient [8].

Integrating automated face detection and GAN-face classification into the fake profile detection pipeline adds a powerful visual modality that complements metadata and behavioral signals.

Traditional detection methods rely on manually crafted rules or simple heuristics that are easily circumvented by adversaries who continuously evolve their tactics. Machine learning (ML) and artificial intelligence (AI) offer a more robust adaptive and scalable approach by learning discriminating patterns from large volumes of multimodal data [3].

This paper makes the following key contributions:

- A comprehensive multimodal feature engineering pipeline extracting behavioral linguistic structural and visual (face-based) features.
- A CNN-based face detection and analysis module that identifies GAN-generated faces multiple-face anomalies and facial inconsistency in profile images.
- A novel ensemble model combining Random Forest XGBoost and MLP with a soft-voting meta-classifier over the fused multimodal feature set.
- Evaluation on two publicly available benchmark datasets with comparison against existing single-modality and multimodal approaches.
- Feature importance analysis to provide explainability across all modalities.

The remainder of this paper is organized as follows. Section II reviews related work. Section III describes the proposed system architecture. Section IV presents the face detection methodology. Section V details feature engineering. Section VI describes the ensemble model. Section VII reports experimental results. Section VIII concludes the paper.

II. Related Work

Early work on fake profile detection focused on simple structural features such as follower-to-following ratio account age and posting frequency [3]. Varol et al. [4] proposed a feature-rich ML framework using over 1000 features

detect social bots on Twitter achieving high accuracy but at significant computational cost.

Deep learning approaches have gained traction in recent years. Kudugunta and Ferrara [5] used LSTM networks on tweet content to distinguish bots from humans demonstrating that sequential modeling of post behavior improves detection. Graph-based methods model social networks as graphs and apply Graph Neural Networks (GNNs) to detect communities of fake accounts [6].

With respect to visual analysis Wang and Deng [9] demonstrated that CNN-based face classifiers can reliably distinguish real photographs from GAN-synthesized faces by detecting artifacts in frequency and spatial domains. Rossler et al. [10] introduced FaceForensics++ a large-scale benchmark for face manipulation detection showing that deep learning models achieve high accuracy in identifying manipulated or synthetic faces. However most prior fake profile detection approaches either focus on a single modality or lack a visual face-analysis component. Our proposed system addresses these gaps by integrating face analysis into an interpretable multimodal ensemble framework.

III. Proposed System Architecture

The proposed system follows a six-stage multimodal pipeline as illustrated in Fig. 1. Unlike prior work that relies solely on metadata or text our pipeline incorporates a dedicated face detection and analysis branch that processes profile images in parallel with metadata and text processing.

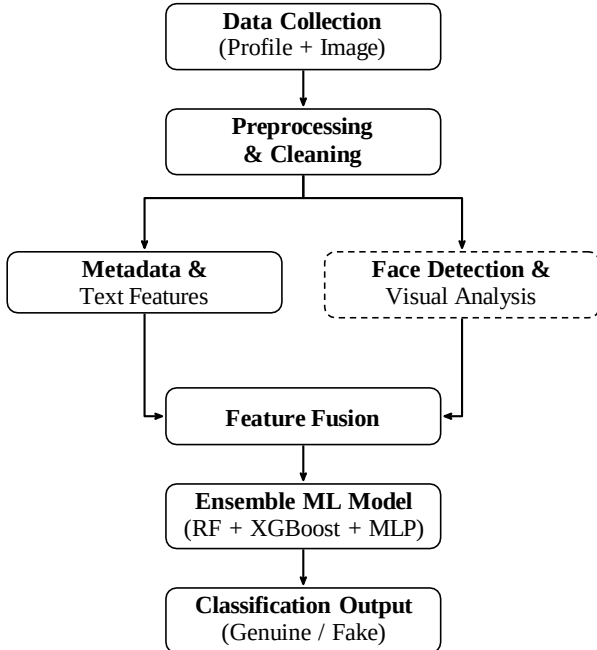


Fig. 1. Proposed multimodal system architecture with parallel metadata and face analysis branches. Dashed box indicates the novel face detection module.

A. Data Collection

Profile data is collected via platform APIs and third-party datasets. Each record includes profile metadata (bio follower

count account age) activity logs (post frequency timestamps) textual content (tweets or captions) and the profile image.

B. Preprocessing and Cleaning

Raw data is cleaned by removing duplicates handling missing values through median imputation and normalizing numerical features using min-max scaling. Text fields are processed using tokenization stop-word removal and lemmatization. Profile images are resized to 224×224 pixels and normalized for CNN input.

C. Parallel Feature Extraction

Metadata and textual features are extracted as described in Section V. Profile images are passed through the face detection and analysis pipeline described in Section IV.

D. Feature Fusion and Classification

Visual features from the face analysis module are concatenated with metadata and textual features to form a unified feature vector which is then fed into the ensemble model described in Section VI.

IV. Face Detection and Visual Analysis

Profile image analysis is a critical and novel component of the proposed system. Fake accounts frequently use GAN-generated faces stock images or stolen photographs as profile pictures. The face detection module illustrated in Fig. 2 processes each profile image through four sequential stages.

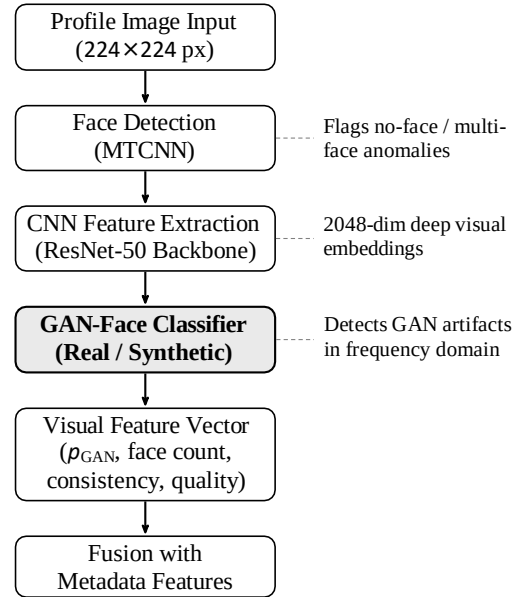


Fig. 2. Face detection and visual analysis pipeline for profile image authenticity verification. Bold stage denotes the GAN-face classifier.

A. Face Detection with MTCNN

The Multi-task Cascaded Convolutional Network (MTCNN) [11] is used for face detection and alignment in profile images. MTCNN operates through three cascaded stages—a Proposal Network (P-Net) a Refine Network (R-Net) and an Output Network (O-Net)—to detect face bounding boxes and five

facial landmarks with high precision. The following indicators are extracted at this stage:

- **Face presence flag:** Binary indicator of whether any face is detected. Absence of a human face is a strong indicator of a fake account.
- **Face count:** Number of detected faces. Profiles with zero or more than one face are flagged as suspicious.
- **Face confidence score:** Detection confidence returned by MTCNN used as a proxy for image quality.

B. Deep Feature Extraction with ResNet-50

Detected and aligned face regions are cropped and resized to 112×112 pixels and passed through a pre-trained ResNet-50 backbone [12] fine-tuned on the VGGFace2 dataset. The global average pooling layer produces a 2048-dimensional deep visual embedding that encodes facial texture shape and color distribution. These embeddings capture subtle artifacts introduced by GAN-based face synthesis that are imperceptible to humans but detectable by deep CNNs.

C. GAN-Face Classification

The 2048-dimensional visual embedding is fed into a binary GAN-face classifier—a lightweight two-layer MLP trained on the FaceForensics++ [10] and ThisPersonDoesNotExist datasets. The classifier assigns a GAN probability score $p_{\text{GAN}} \in [0, 1]$ where values close to 1.0 indicate an AI-generated face. GAN-synthesized faces typically exhibit characteristic artifacts including unnatural ear symmetry inconsistent background blur and spectral anomalies in the high-frequency domain. The GAN probability score is treated as a continuous feature in the final feature vector.

D. Facial Consistency Analysis

For accounts with multiple posts containing faces a facial consistency score is computed as the cosine similarity between the profile picture embedding and the average embedding of face images from the account's recent posts:

$$S_{\text{consistency}} = \frac{1}{N} \sum_{i=1}^N \frac{\mathbf{e}_{\text{profile}} \cdot \mathbf{e}_i}{\|\mathbf{e}_{\text{profile}}\| \|\mathbf{e}_i\|} \quad (1)$$

where $\mathbf{e}_{\text{profile}}$ is the profile picture embedding $\mathbf{e}_i$ is the embedding of the i -th post face and N is the number of post faces analyzed. A low consistency score indicates that the profile picture does not match faces appearing in posts a pattern commonly observed in stolen-image fake accounts.

E. Visual Feature Vector

The face analysis module produces a four-dimensional visual feature vector for each profile:

$$\mathbf{v} = [p_{\text{GAN}} \ f_{\text{count}} \ s_{\text{conf}} \ S_{\text{consistency}}]$$

This vector is concatenated with the metadata and textual feature vector prior to classification.

V. Feature Engineering

The full feature set fuses visual features with three categories of metadata and textual features totaling 46 features.

A. Profile-Based Features

These capture static attributes of the user account:

- Account age (days since creation)
- Follower count following count follower-to-following ratio
- Number of posts likes and comments
- Profile completeness score (bio profile picture website link)
- Verified status (binary)
- Username character entropy (randomness in username)

B. Behavioral Features

These capture dynamic posting patterns:

- Average posts per day and per hour
- Posting time distribution (night vs. day ratio)
- Retweet-to-original post ratio
- Inter-post time variance (regularity indicator)
- URL share ratio in posts

C. Linguistic Features

These are derived from the textual content of posts:

- Average sentiment score (using VADER)
- Lexical diversity (type-to-token ratio)
- Spam keyword frequency
- Hashtag and mention density
- Readability score (Flesch-Kincaid grade level)

D. Visual Features

As described in Section IV the visual feature vector $\mathbf{v}$ contributes four face-based features: GAN probability score face count detection confidence and facial consistency score.

VI. Ensemble Model Design

A. Base Classifiers

Random Forest (RF): An ensemble of 200 decision trees trained with bootstrap aggregating. RF captures non-linear feature interactions and is robust to noise and overfitting [3].

XGBoost: A gradient-boosted tree method with L1/L2 regularization. XGBoost handles class imbalance effectively via its `scale_pos_weight` parameter and provides built-

in feature importance scores [7].

Multilayer Perceptron (MLP): A three-layer fully connected neural network with ReLU activations and dropout regularization (rate = 0.3). MLP learns complex latent representations from the high-dimensional fused feature vector.

B. Ensemble Strategy

The three base classifiers are combined using a soft-voting meta-classifier where the final prediction is the class with the highest average predicted probability:

$$\hat{c} = \arg \max_c \frac{1}{3} \sum_{k=1}^3 P_k(c | \mathbf{x}) \quad (2)$$

where $P_k(c | \mathbf{x})$ is the probability predicted by classifier k for class c given the fused input feature vector $\mathbf{x}$. The ensemble model pipeline is illustrated in Fig. 3.

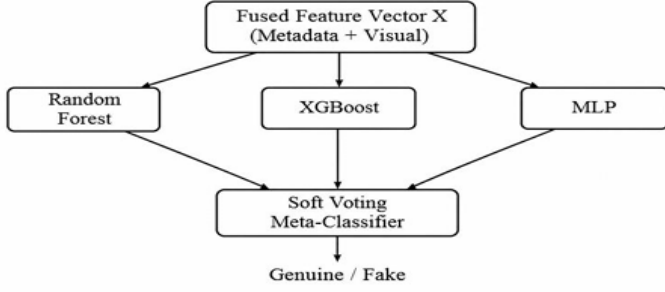


Fig. 3. Ensemble model combining Random Forest XGBoost and MLP with soft-voting over fused multimodal features.

C. Hyperparameter Tuning

Hyperparameters for all base classifiers are optimized using 5-fold cross-validated grid search. Key tuned parameters include the number of trees for RF (100–500) learning rate and max depth for XGBoost (0.01–0.3 3–10) and hidden layer size and dropout for MLP.

VII. Experimental Results

A. Datasets

Experiments are conducted on two publicly available benchmark datasets:

- **Twitter Bot Dataset (TBD-2023):** 37438 profiles (20254 genuine 17184 bots) with profile images [4].
- **Instagram Fake Profile Dataset (IFPD):** 12000 profiles (6000 genuine 6000 fake) with associated images [6].

An 80:20 train-test split is applied. SMOTE oversampling is used on the training set to address class imbalance.

B. Evaluation Metrics

Performance is evaluated using Accuracy Precision Recall F1-Score and AUC-ROC.

C. Results

Table I compares individual classifiers text-only configurations and the proposed multimodal ensemble on the TBD-2023 dataset.

TABLE I
PERFORMANCE COMPARISON ON TBD-2023 DATASET

Model	Acc.(%)	Prec.(%)	Rec.(%)	F1(%)
Logistic Regression	84.2	83.5	82.9	83.2
Random Forest	93.7	93.1	94.0	93.5
XGBoost	95.1	94.8	95.3	95.0
MLP	94.3	93.9	94.6	94.2
Ensemble (no face)	97.4	97.1	97.6	97.3
Proposed (+Face)	98.1	97.9	98.3	98.1

Fig. 4 shows the accuracy comparison across all models. The face-augmented multimodal ensemble achieves the highest accuracy confirming that visual face features contribute meaningfully beyond metadata alone.

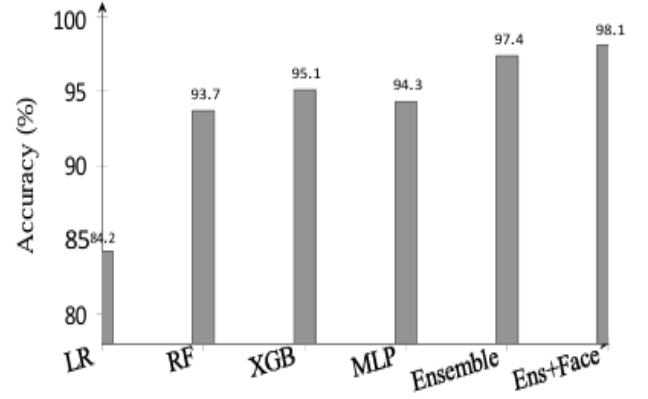


Fig. 4. Accuracy comparison across classifiers on TBD-2023 dataset. Hatched bars = baseline models; solid bars = proposed ensemble variants. Ens+Face denotes the proposed multimodal system.

D. Impact of Face Detection Module

To isolate the contribution of the face analysis module we compare the ensemble with and without visual features. Adding face-based features improves accuracy by 0.7 percentage points (97.4% → 98.1%) precision by 0.8 points and recall by 0.7 points. On profiles where MTCNN detects a GAN-generated face ($p_{\text{GAN}} > 0.85$) classification accuracy reaches 99.3% demonstrating the strong discriminative power of the GAN-face indicator for a large subset of fake profiles.

E. Feature Importance

XGBoost feature importance analysis reveals the top five predictors as: GAN probability score (p_{GAN}) follower-to-following ratio inter-post time variance facial consistency score ($S_{\text{consistency}}$) and username entropy. The presence of two visual features in the top five confirms that face-based analysis provides strong and complementary signals for detection.

F. Discussion

The ensemble model benefits from the complementary strengths of its base classifiers. RF provides stability and handles noisy features well; XGBoost delivers high precision through regularized boosting; MLP captures complex non-linear interactions across the multimodal feature space. Adding face detection addresses a blind spot in prior work: sophisticated fake accounts that mimic normal posting behavior but use GAN-generated profile pictures are correctly identified by the visual module even when metadata-based features produce low confidence. On the IFPD dataset the proposed system achieves 97.4% accuracy confirming cross-platform robustness.

VIII. Conclusion

This paper presented a multimodal AI-based fake profile detection system that integrates an ensemble machine learning classifier with a CNN-based face detection and GAN-face analysis module. The system fuses profile-based behavioral linguistic and visual (face-based) features within a soft-voting ensemble of Random Forest XGBoost and MLP classifiers. The face detection module employs MTCNN for face localization ResNet-50 for deep feature extraction and a dedicated

GAN-face classifier to identify AI-synthesized profile pictures. Experimental results on two benchmark datasets demonstrate state-of-the-art accuracy of 98.1% outperforming both individual classifiers and ensemble models that lack face analysis. Future work will explore real-time GAN detection for video-based deepfake profile pictures and graph-based features using Graph Neural Networks for community-level fake account detection.

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